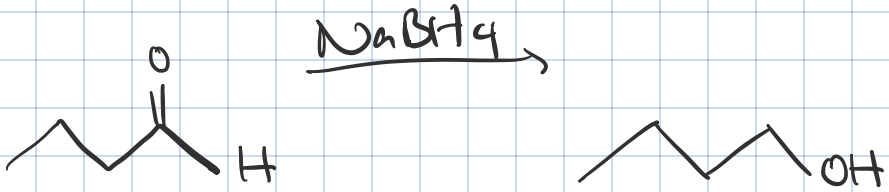
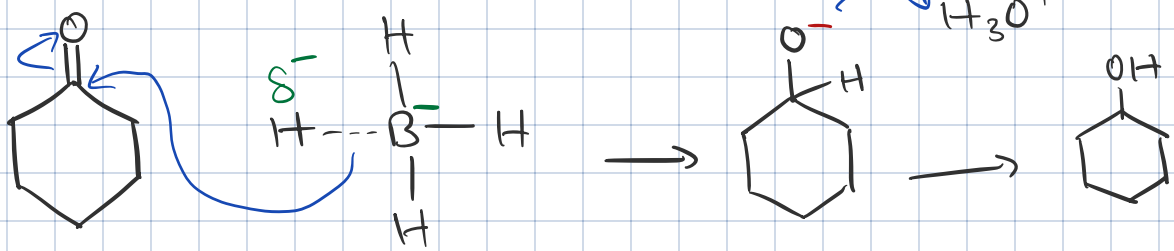
$NaBH_4$: mild reducing agent, adds one hydrogen atom

$LiAlH_4$: strong

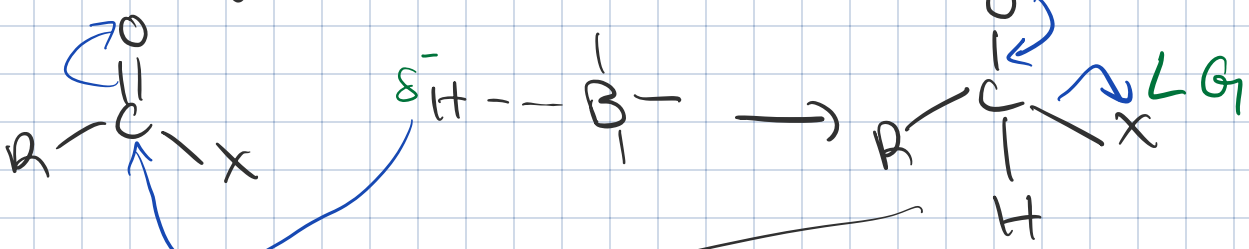


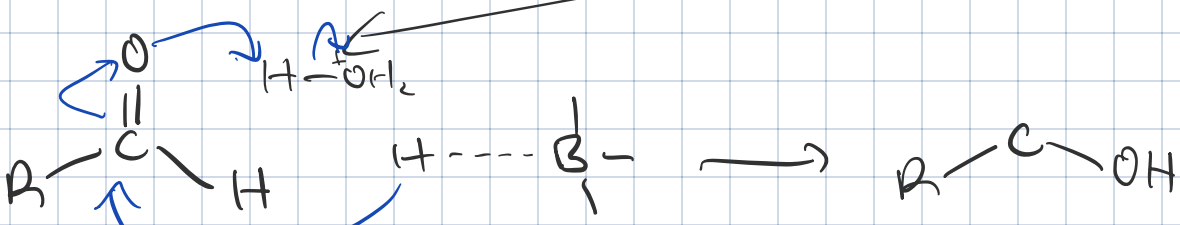
Diff Mech

Mech:

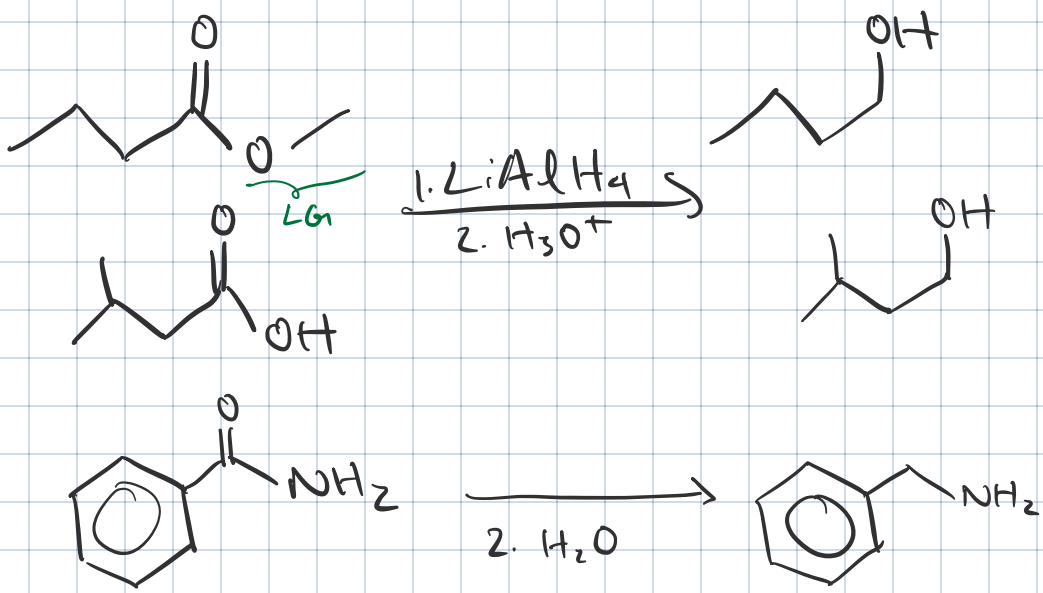


Mech acyl halide



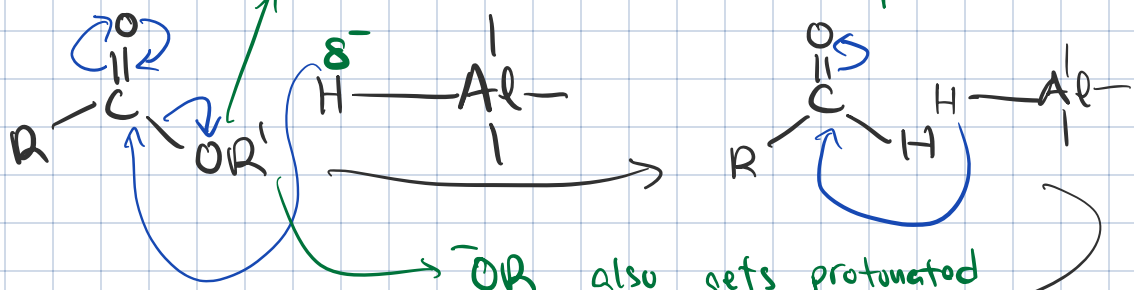


$LiAlH_4$: reduces anything



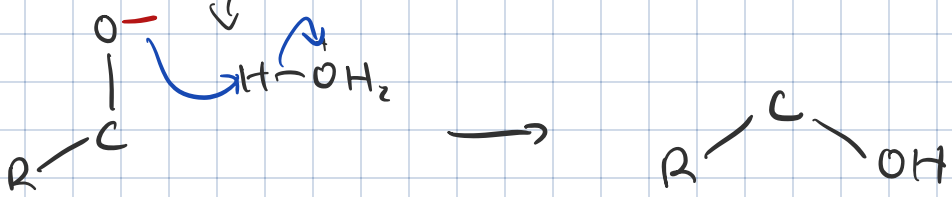
Best LG out of the 3 possible

1.



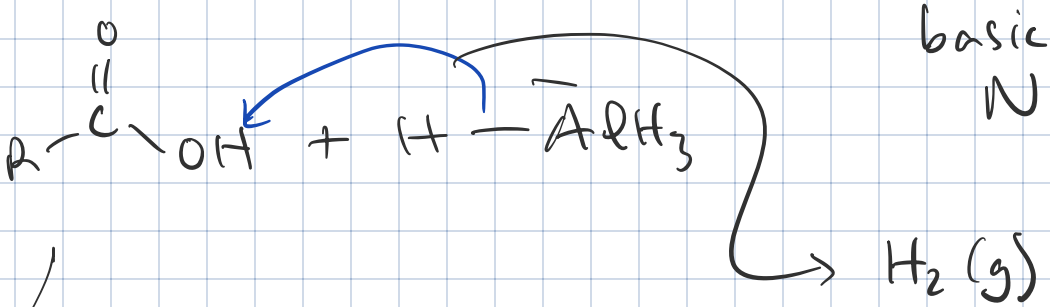
^-OR also gets protonated in step 2

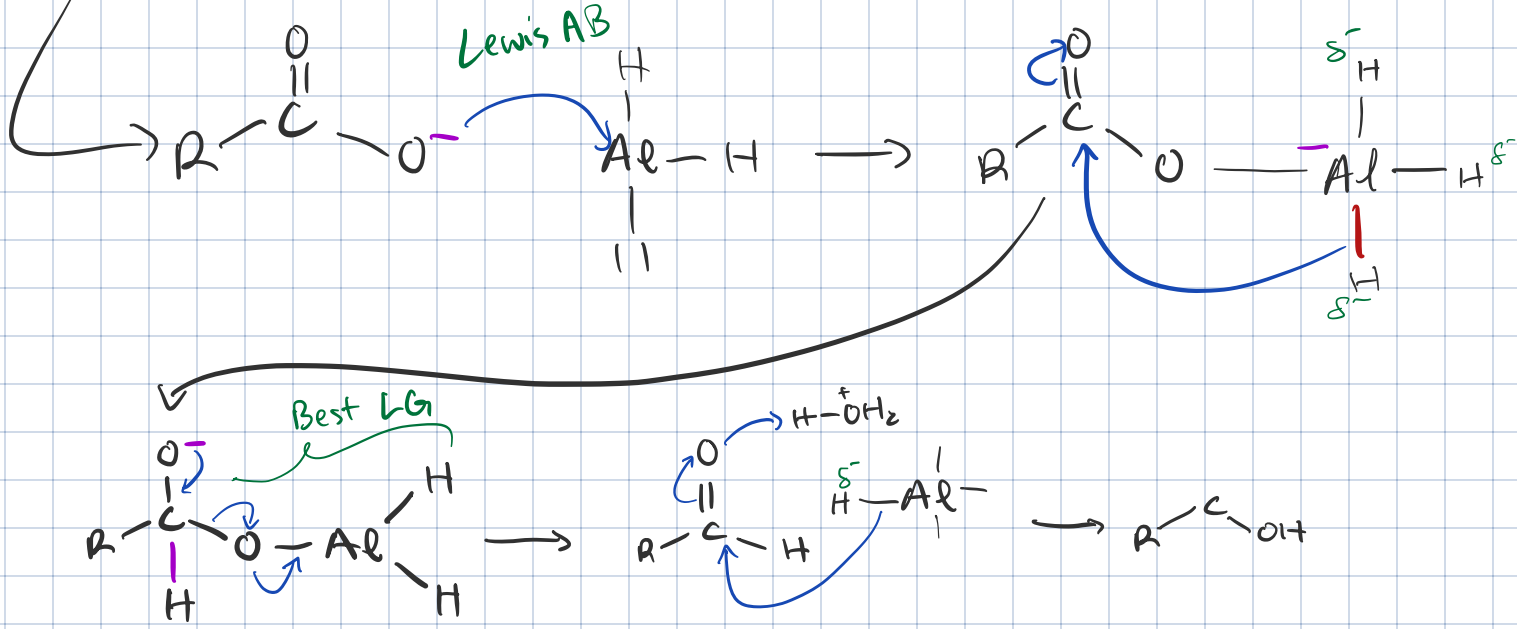
2.



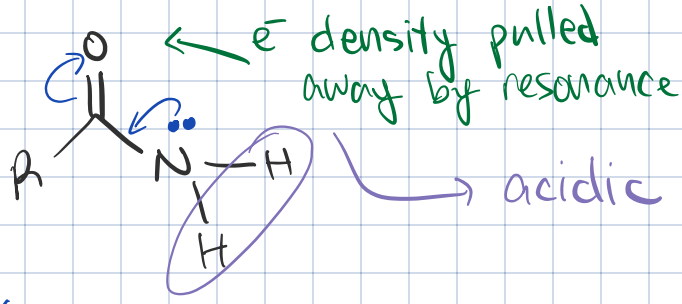
Mech for Carboxylic Acid

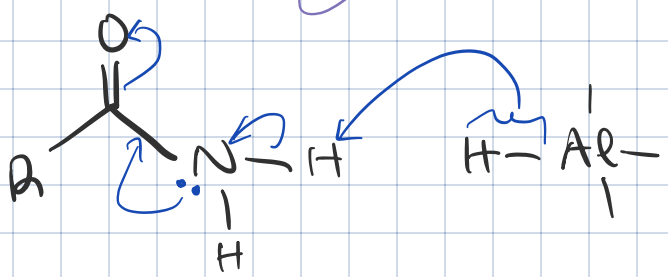
AlH_4^- is more basic than NaBH



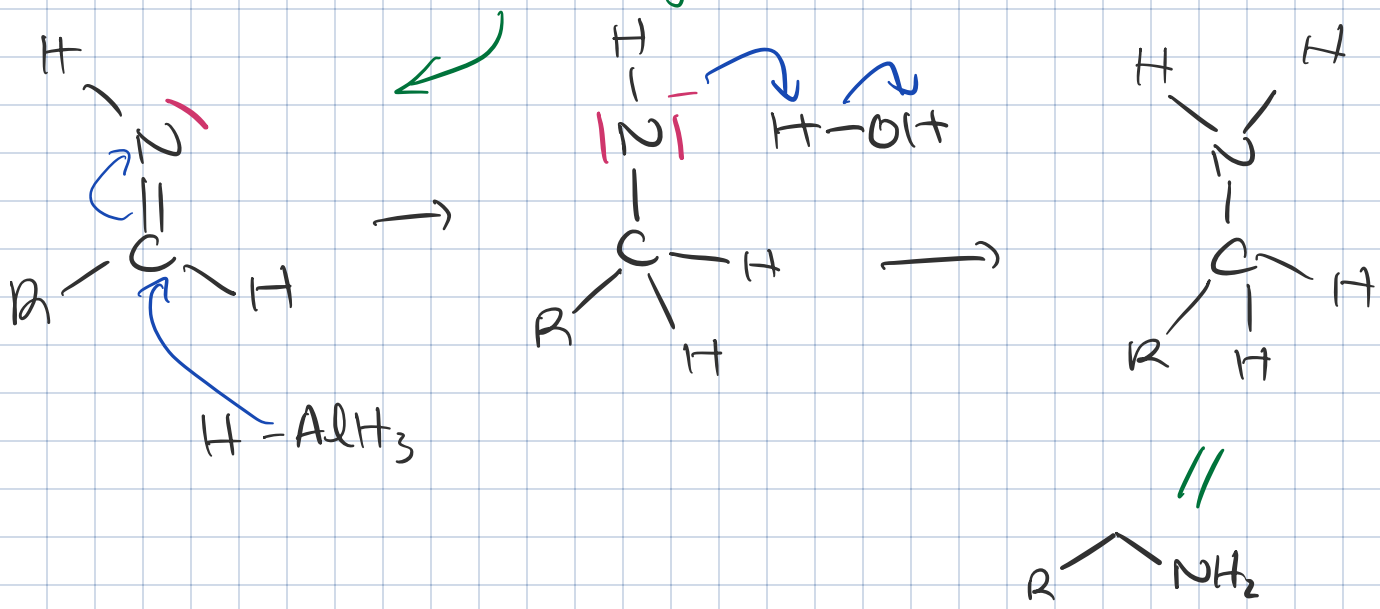


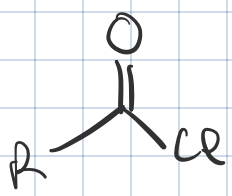
Reduction of amides





Similar to carboxylate reduction above



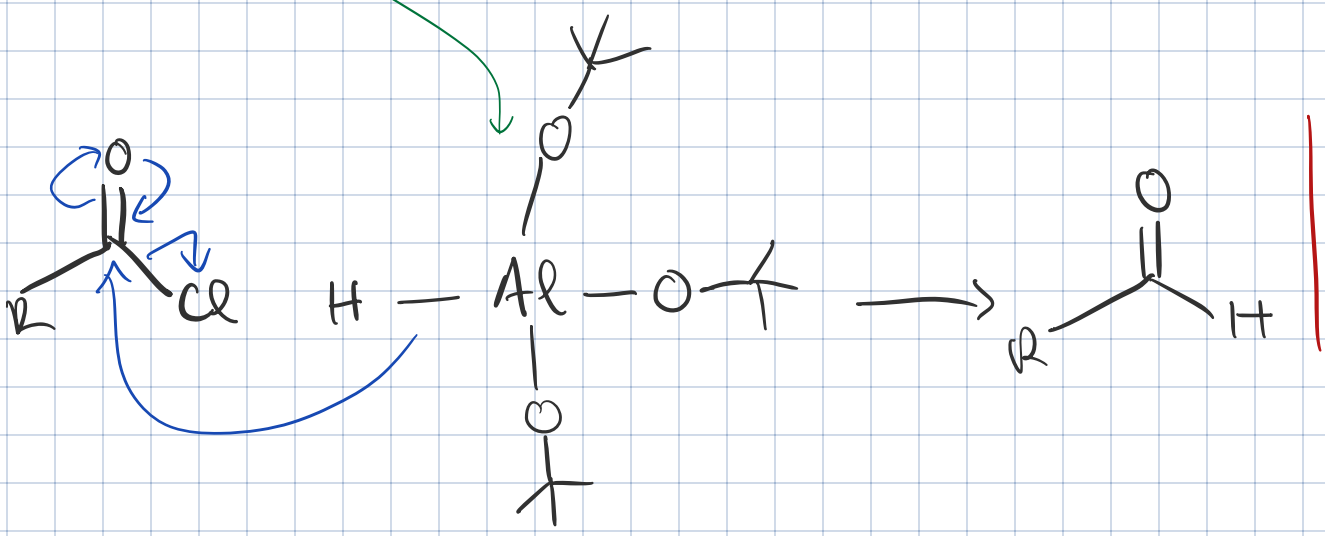


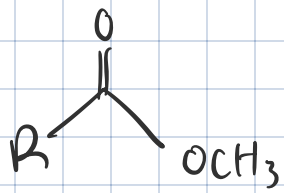
$\xrightarrow[2.\ H_2O]{1.\ LiAl[OC(CH_3)_3]_3H}$

only 1 Hydride, stops rxn after 1st reduction

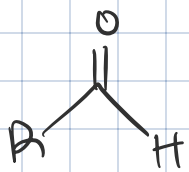
acyl halide $\xrightarrow{red}$ aldehyde ~~→~~ alcohol

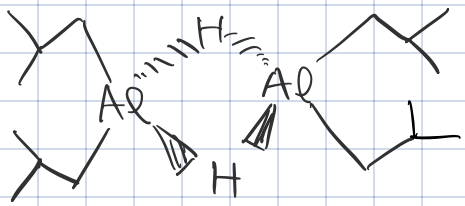
stopped





$\xrightarrow[2.\ \Delta]{1.\ DiBAl\text{-}H}$ (−78°C)



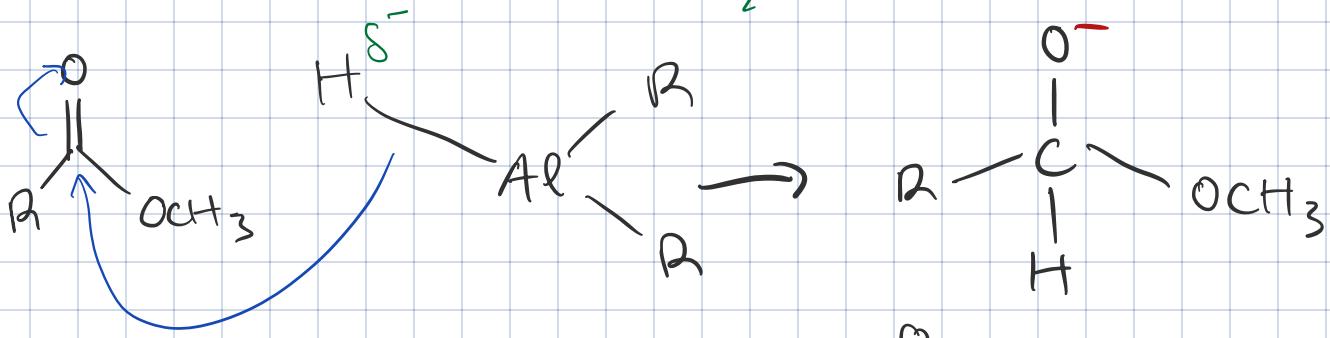


1 H^- per Dibal

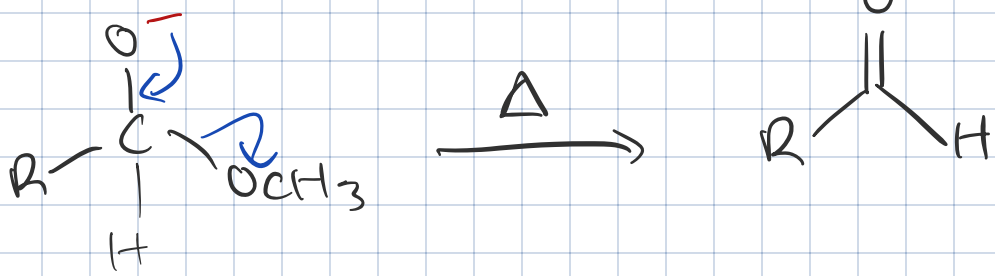
simplified

Mech:

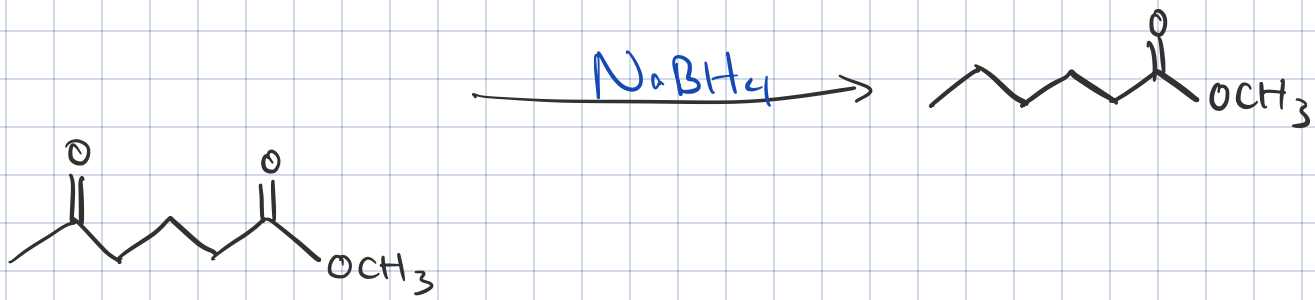
1.

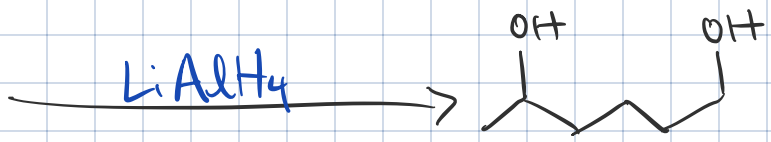


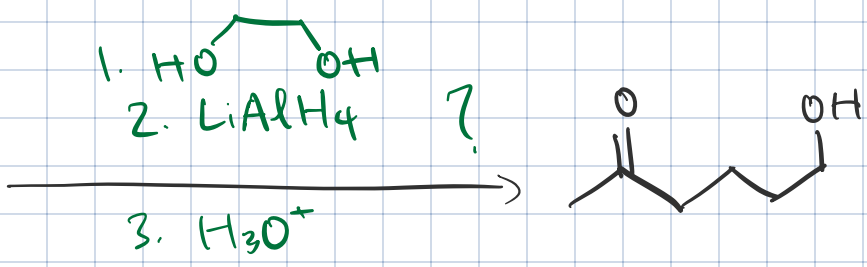
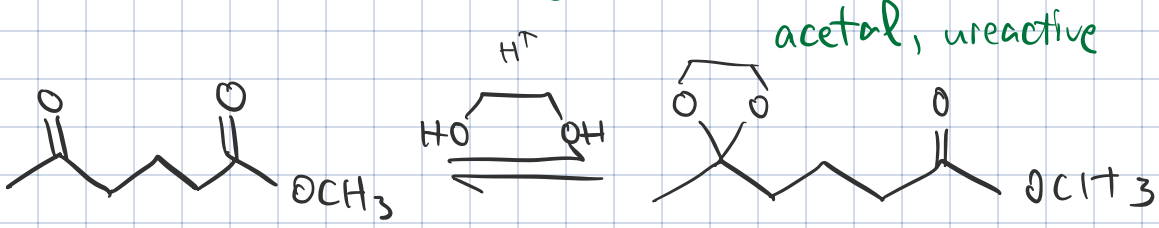
2.



Review







Protecting Group: